

Joao Catarino [Joe-wow]

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EXPERIENCE

• J.P. Morgan [🌐]

London, UK

Graduate - Associate - Senior Associate

September 2020 - January 2023 - January 2024 - Present

- Served as a Front Office Engineer in Buy-Side Equities, engineering core backend and data services for the Asset Management Investment Platform. The platform, built on a microservices architecture using Python, Java, Cassandra, S3 and more, processes 1M+ daily requests and 30TB of financial data across hybrid cloud infrastructure.
- Partnered directly with Portfolio Managers and Quantitative Researchers to productionize ML models, alpha signals, risk metrics, and portfolio analytics for Equities desks. Strong impact led to current expansion to Global Fixed Income, Currency and Commodities.
- Led the development of a highly extensible low/no code Python ETL framework with 100+ modules, now used in 800+ pipelines daily. Reduced time to market of new signals from weeks to days, while providing cost savings, modernizing the technology stack and enhancing start-of-day resiliency.
- Designed and deployed a full production-grade Airflow environment for orchestration of data pipelines.
- Pioneered infrastructure as code, containerization and event-driven pipelines to improve team efficiency.
- Integrated multi-source financial data feeds from Bloomberg, FactSet, MSCI, and proprietary feeds.

Software Engineering Intern

Glasgow and London, UK

Summer 2018 and Summer 2019

- Engineered a service to facilitate app-to-app authentication and created a proof of concept for modernizing the infrastructure to the cloud for the Portfolio Implementation team.
- Worked on the pricing, risk and trade management platform for Rates and Credit, in Python.

• Seedrs (now Republic Europe) [🌐]

Lisbon, Portugal

Data Science Intern

Summer 2017

- Developed a recommendation system for the marketplace using pandas, numpy and scikit-learn.

EDUCATION

• The University of Edinburgh [🌐]

Edinburgh, UK

Master of Informatics

September 2015 - July 2020

- First Class with Honours. System Design Project runner-up. Awarded the Edinburgh Award.
- Led student mentoring sessions and assisted in organizing the university's largest hackathon.
- Distributed Systems, Distributed Ledgers, Databases, Machine Learning and Data Science courses.

LEADERSHIP

• Edinburgh University Trading and Investment Club [🌐]

Edinburgh, UK

Head of IT

2018 - 2020

- Responsible for all digital footprint of EUTIC and the Edinburgh Finance Conference.

Head of TMT Sector

2017

- Tripled member attendance and investment pitches.
- Best Performing Sector Prize and TMT members awarded Best Pitch Prize.

PROJECTS

• Rhezus

Tools: Flask, Angular, DynamoDB, AWS Lambdas, Elastic Beanstalk

- Built a web-based medical education tool used by 900+ students to prepare for the final national exam.

• Clave

Tools: Go, Swift, DynamoDB

- Co-developed an iOS app for geofenced anonymous discussions.

SKILLS

- **Certifications:** AWS Associate Solutions Architect (2022), CFA Institute Investment Foundations (2021)
- **Backend:** Airflow, Pandas, Numpy, Pydantic, SQL, Solidity
- **DevOps:** Docker, Kubernetes, AWS ECS, SQS, Terraform, Jenkins, Git